

AMENDMENT NO. 1 MARCH 2026
TO
**IS 9500 : 2017 360 DEGREE ROTARY EMERGENCY WARNING LIGHTS/
SPECIAL WARNING LAMPS — SPECIFICATION**
(First Revision)

(Page 1, clause 2) — Substitute the following for the existing:

‘2 REFERENCES

The standards given below contains provision, which through reference in this text, constitutes provision of this standard. At the time of publication the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent edition of these standards:

<i>IS No.</i>	<i>Title</i>
IS 14272 : 2011	Automotive vehicles — Types — Terminology (<i>first revision</i>)
IS 18606 : 2024	Electric power train of M and N category vehicles — Specific requirements’

(Page 1, clause 3.1.3) — Insert the following at the end:

‘3.1.4 Half Bar — A special warning lamp with one or more optical systems emitting light intermittently from 135° left to 135° right relative to its horizontal reference axis, which is intended to be mounted on the vehicle either to the front or to the rear of the vehicle.’

(Page 1, clause 3.2) — Substitute the following for the existing:

‘3.2 Special Warning Lamps of Different Types

Special warning lamps which differ intrinsically in such matters as:

- a) Lamps bearing the same trade name or mark but produced by different manufacturers are considered as being of different types;
- b) Lamps produced by the same manufacturer differing only by the trade name or mark may be considered to be of the same type;
- c) Size and form of the coloured cover;
- d) The optical system;
- e) Nature of the beam (for example, rotating or stationary flashing);
- f) The colour of the light emitted;
- g) The colour and/or type of the light source;
- h) The colour and construction of the light source module; and
- j) Whether the special warning lamp has one level (class 1) or two levels (class 2) of intensity.’

(Page 1, clause 3.4) — Substitute the following for the existing:

“3.4 ‘On’ Time t_H

It is the period of time within which the luminous intensity of the flashing light is superior to 1/10 of the maximum value (peak value) J_m . In case of groups of several flashes the ‘on’ time shall be measured from the beginning of the first flash of the group to the end of the last flash of the same group.”

Price Group 2

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(Page 2, clause 3.7) — Substitute the following for the existing:

‘3.7 Reference Centre of the Special Warning Lamp

For a rotating or stationary flashing lamp (Category T) and for a directional flashing lamp (Category X), the intersection of the axis of reference with the exterior light-emitting surface. It is specified by the manufacturer of the special warning lamp. In the absence of such specification, it means:

- a) The optical centre of the light source; or
- b) The geometric centre of the external optical surface; or
- c) In case of an array of light sources in the optical system, the geometric centre of the array; shall be considered as the reference centre.’

(Page 2, clause 3.8) — Substitute the following for the existing:

‘3.8 Reference Axis of the Special Warning Lamp

- a) For a rotating or stationary flashing lamp (Category T), a vertical axis passing through the reference centre of the lamp;
- b) For a directional flashing lamp (Category X) or a half bar (Category HT), a horizontal axis parallel to the median longitudinal plane of the vehicle; and
- c) The manufacturer of the special warning lamp shall indicate the position of the special warning lamp in relation to the reference axis.’

(Page 2, clause 3.9.1.2) — Insert the following at the end:

‘3.9.1.3 In cones, the generating lines of which produce with the above-mentioned horizontal plane angles, starting at a point where the effective intensity is minimum, the values which are indicated in the table of Annex D.’

(Page 3, clause 6.1) — Substitute the following for the existing:

‘6.1 The special warning lamps shall be so designed and constructed that:

- a) In normal conditions of use and notwithstanding the vibrations to which they may be subjected in such use, their satisfactory operation remains assured and they retain the characteristics prescribed in this standard; and
- b) The relevant requirements with regard to internal voltage higher than 60 V d.c. are fulfilled, for example by marking the device, as defined in 5.1.1.4 of IS 18606.’

(Page 3, clause 6.7) — Insert the following at the end:

‘6.8 A rotating or flashing special warning lamp device of Category T may emit light of several colours. In this case, all the requirements shall be met for each colour separately over the full angular range specified.

The activation of more than one colour at the same time shall be prohibited.

The lamp manufacturer shall supply mounting information, for correct mounting on a vehicle, to ensure that only one colour of the special warning lamp is activated at the same time.

6.9 For approved special warning lamps, it shall be not possible for the user to activate groups of several flashes (flash patterns), which do not conform to the requirements in **D-2.5**.

6.10 In case of a magnetic attachment the special warning lamp shall be exposed to the test described in Annex F. During the test the magnetic base mounting shall not move by more than 200 mm from the original position.

NOTE — The manufacturer shall inform the user, that an attachment is only possible for a roof of adequate steel or a steel mounting plate.’

(Page 5, clause C-2) — Substitute the following for the existing:

‘C-2 During the test, the device shall turn on its vertical axis at a rate of 4 turns/min. However, if the water is simultaneously directed to the device under test from all directions in the horizontal plane using a multitude of nozzles, there is no need to rotate the device during the test. In this latter case the water flow specified above shall be adjusted accordingly to achieve even distribution and the correct precipitation.’

(Page 5, clause D-1.1) — Substitute the following for the existing:

‘D-1.1 Measurements of the photometric characteristics shall be taken at a distance of at least 25 m. The angular diameter of the photoelectric receiver as seen from the special warning lamp shall be 10 min of arc maximum. However, the distance of the sensor from the special warning lamp should be adjusted to a longer distance, such that the aperture through which the sensor is receiving the light allows full view of the special warning lamp for the sensor.’

(Page 5, clause D-1.2) — Substitute the following for the existing:

‘D-1.2 For special warning lamps having one level of intensity (class 1), the ‘by night’ level shall apply. For special warning lamps having two levels of intensity (class 2) measurements shall be carried out for each of the two levels. The effective luminous intensities in various directions shall be as specified in the D-3.2 and D-3.3 and shall be measured after the light output from the special warning lamp has reached photometric stability as specified in D-2.4.’

(Page 5, clause D-2.4) — Substitute the following for the existing:

‘D-2.4 For any lamp, the luminous intensities measured after one minute and after the light output from the special warning lamp has reached photometric stability (deviation of less than ± 5 percent in the last 15 min of operation) shall comply with the minimum and maximum requirements. The luminous intensity distribution after one minute of operation can be calculated by applying the ratio achieved at HV between one minute and at photometric stability.’

(Page 7, clause D-4.2) — Substitute the following for the existing:

‘D-4.2 If a special warning lamp contains two or more optical systems, all the optical systems shall work in phase within each half of a complete ‘bar’ which is designed to extend on the width of the vehicle. In such a case, for the purpose of measurement of effective intensity, only one half of the ‘bar’ shall be energized so that the light emission from the side not being measured is not added into the side being measured. The timing measurements as described in D-2.5 apply to the operating half of the ‘bar’.’

(Page 8, Annex E) — Insert the following at the end:

‘ANNEX F

(Clause 6.10)

TEST FOR THE BASE MOUNTING OF MAGNETICALLY ATTACHED SPECIAL WARNING LAMPS

F-1 The tests shall be performed at $23\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$ (ambient temperature) on a metal surface (refer Fig. 1) of sufficient size and the following specifications:

- a) Type of metal: steel with a nominal yield strength of 180 N/mm² to 240 N/mm²;
- b) Thickness of metal surface: $0.7^{0.1}_{0.0}$ mm;
- c) Radius of curvature: < 5 000 mm;
- d) Paint thickness: 120 micrometres $\pm$ 20 micrometres;
- e) Paint protection: polyurethane film;
- f) Paint protection thickness: > 200 micrometres including glue; and
- g) The metal surface shall be > 20 mm above any other ferrous or magnetic material.

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FIG. 1 METAL SURFACE WITH PAINT AND PAINT PROTECTION FILM

F-2 The special warning lamp shall be magnetically attached to the flat metal surface and subjected to a single shock pulse with a minimum acceleration of 16 g during 30 ms. The direction of the acceleration shall be in horizontal direction.'